

Instructions to the Students:

- 1. All the questions are compulsory.*
- 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.*
- 3. Use of non-programmable scientific calculators is allowed.*
- 4. Assume suitable data wherever necessary and mention it clearly.*

	(Level/CO)	Marks
Q.1 Solve Any Two of the following.		12
A) Explain the following types of pulleys with neat sketch ,	CO1	6
i. idler pulleys		
ii. intermediate pulleys		
iii. loose & fast pulleys		
iv. guide pulleys		
B) What are different modes of transmitting power from one shaft to another? Compare them.	CO1	6
C) A 100 mm & 10 mm thick belt transmits 5 kW of power between two parallel shafts. The distance between shaft centers is 1.5 m & diameter of the smaller pulley is 440 mm. The driving & driven shafts rotate at 60 rpm & 150 rpm respectively. The coefficient of friction is 0.22. Find the stress in the belt if the two pulleys are connected by	CO1	6
i. An open belt; and		
ii. A cross belt		
Q.2 Solve Any Two of the following.		12
A) Give a detailed classification of Gears.	CO2	6
B) Define Circular pitch, Addendum, dedendum, pitch circle diameter, module & backlash.	CO2	6
C) Describe the features of involute tooth profile in detail.	CO2	6
Q.3 Solve Any Two of the following.		12
A) What is the difference between a simple gear train & compound gear train? Explain with the help of sketches.	CO2	6

- B) An epicyclic gear train consist of an arm & two gears A & B having 30 & 40 teeth respectively. The arm rotates about the centre of the gear A at a speed of 80 rpm counter clockwise. Determine the speed of the gear B if,
- the gear A is fixed, and
 - the gear A revolves at 240 rpm clockwise instead of being fixed

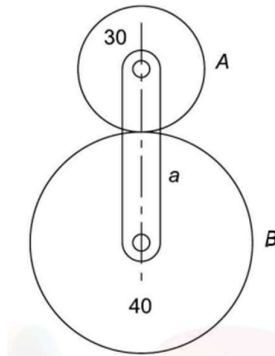


Fig.1

- C) Describe the construction & working of sliding gear box with neat sketch. CO2 6

Q.4 Solve Any Two of the following. 12

- A) In a Hartnell governor, the extreme radii of rotation of the balls are 40 mm & 60 mm; and the corresponding speeds are 210 rpm & 230 rpm. The mass of each ball is 3 kg. The lengths of the ball & the sleeve arms are equal. Determine the initial compression & the constant of the central spring. CO3 6
- B) Describe the role of flywheel & governor in an automobile. CO4 6
- C) An aero-plane flying at 240 km/h turns towards left & completes a quarter circle of 60 m radius. The mass of rotary engine & the propeller of the plane is 450 kg with a radius of gyration of 320 mm. The engine speed is 2000 rpm clockwise when viewed from the rear. Determine the gyroscopic couple on the aircraft & state its effect. CO5 6

In what way is the effect changed when the

- Aeroplane turns towards right
- Engine rotates clockwise when viewed from the front (nose end) & the aeroplane turns (a) left (b) right?

Q.5 Solve Any Two of the following. 12

- A) What is vibration? What are the various types of vibration? CO6 6
- B) Find the natural frequency of vibration of the system shown in Fig.2. (K₁= 1000 N/m, K₂ = 1000 N/m, K₃ = 2000 N/m, m =10 kg) CO6 6

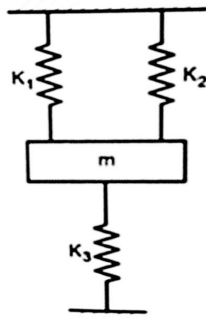


Fig.2.

C) Describe types of dampings in detail.

CO6

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***** End *****